5.1 DIODE IDENTIFICATION

MIKROELEKTRONIKA

European diodes are marked using two or three letters and a number. The first letter is used to identify the material used in manufacturing the component (A – germanium, B – silicon), or, in case of letter Z, a Zener diode.

The second and third letters specify the type and usage of the diode. Some of the varities are:
A – low power diode, like the AA111, AA113, AA121, etc. – they are used in the detector of a radio receiver; BA124, BA125 : varicap diodes used instead of variable capacitors in receiving devices, oscillators, etc., BAY80, BAY93, etc. – switching diodes used in devices using logic circuits. BA157, BA158, etc. – these are switching diodes with short recovery time.
B – two capacitive (varicap) diodes in the same housing, like BB104, BB105, etc.
Y – regulation diodes, like BY240, BY243, BY244, etc. – these regulation diodes come in a plastic packaging and operate on a maximum current of 0.8A. If there is another Y, the diode is intended for higher current. For example, BYY44 is a diode whose absolute maximum current rating is 1A. When Y is the second letter in a Zener diode mark (ZY10, ZY30, etc.) it means it is intended for higher current.
G, G, PD – different tolerance marks for Zener diodes. Some of these are ZF12 (5% tolerance), ZG18 (10% tolerance), ZPD9.1 (5% tolerance).
The third letter is used to specify a property (high current, for example).
American markings begin with 1N followed by a number, 1N4001, for example (regulating diode), 1N4449 (switching diode), etc.
Japanese style is similar to American, the main difference is that instead of N there is S, 1S241 being one of them.